A per-run bound on the minimiser residual, tightened to the part of the model error the fit cannot absorb

drop-in for Sec. VIII; figure `fig_tight_rms.png`

1 Notation

All quantities refer to one run's post-onset window unless stated. The table is complete; symbols not listed here do not appear.

τ	time since the identified onset, $\tau \in [0, \tau_{\text{end}}]$ [s]
τ_{end}	window length; $x = C_2\tau_{\text{end}}$ is its dimensionless form
$\tau_i,\ N$	the N sample instants $0 = \tau_1 < \cdots < \tau_N = \tau_{\text{end}}$ (logging period $T_s \approx 10$ ms)
y_i	measured roll/pitch rate at τ_i, from the flight IMU [rad/s]
$\omega_{\text{nom}}(\tau)$	nominal response $C_1(\cosh C_2\tau - 1) + C$ [rad/s]
$C_1,\ C_2,\ C$	amplitude $C_1 = K\dot{M}$, exponent $C_2 = \sqrt{Wz_{CoM}/J_P}$, pre-onset baseline C
$K,\ \dot{M}$	calibrated gain $K = 1/(Wz_{CoM})$ [1/(N m s)] and commanded ramp rate [N m/s]
$W,\ z_{CoM},\ J_P$	weight [N], CoM height [m], pivot-axis inertia [kg m^2]; $J_P = 1/(KC_2^2)$
$e_\omega(\tau)$	deviation of the true (noise-free) rate from ω_{nom} [rad/s]
n_i	in-window disturbance: everything in y that is neither ω_{nom} nor e_ω [rad/s]
u_i	regressor $u_i = \cosh C_2\tau_i - 1$; $\tilde{u}_i = u_i - \bar{u}$ its centred form
$V,\ A,\ P$	fitted family's span $V = \text{span}\{u, \mathbf{1}\}$; $A = [\,u \ \ \mathbf{1}\,]$; projector $P = A(A^\top A)^{-1}A^\top$, in closed form $P_{ij} = 1/N + \tilde{u}_i\tilde{u}_j / \sum_k \tilde{u}_k^2$
$\hat{r}$	minimiser residual $\hat{r} = (I - P)\,y$: the per-run fit with C_2 pinned and (C_1, C) free is exactly this projection
$\rho(s)$	deviation forcing at time s: the gravity remainder $Wz_{CoM}(\sin\varphi - \varphi)$ plus the ground-effect terms, evaluated on the true trajectory [N m]
$\bar{\rho}$	a priori pointwise bound $\lvert\rho(s)\rvert \le \bar{\rho}$ from the 10° box, as in (93) [N m]
k_j	kernel column: response of e_ω at the samples to a unit impulse of ρ at τ_j, i.e. $k_j(\tau_i) = \Delta s_j \cosh\big(C_2(\tau_i - \tau_j)\big)/J_P$ for $\tau_i \ge \tau_j$, else 0; Δs_j the trapezoidal quadrature weight [s]
Φ	a priori bound on $\text{RMS}\big((I-P)\,e_\omega\big)$, eq. (T.4) [rad/s, reported in °/s]
$E(\tau)$	the pointwise envelope of (93), $E = \bar{\rho}\sinh(C_2\tau)/(J_P C_2)$; note $E(\tau) = \bar{\rho}\sum_j \lvert k_j(\tau)\rvert$ — the same kernel without the projection
n_{hi}	the part of $\hat{r}$ above $f_c = 5$ Hz; attributable to disturbance because the family and e_ω are smooth on the window scale
κ	spectral shape ratio $\text{RMS}(<f_c)/\text{RMS}(>f_c)$ of a disturbance record
κ_q	κ measured on the run's quiet stretch of equal length, ending at the excitation start
κ_{imp}	in-window implied ratio $\sqrt{\text{RMS}(\hat{r})^2 - \text{RMS}(n_{\text{hi}})^2}\,/\,\text{RMS}(n_{\text{hi}})$
$s_{\text{med}},\ \kappa_b$	median in-window enhancement $s_{\text{med}} = \text{med}(\kappa_{\text{imp}}/\kappa_q)$; campaign constant $\kappa_b = s_{\text{med}}\ \max_{\text{runs}} \kappa_q$
$\text{RMS}(v)$	$\sqrt{\frac{1}{N}\sum_i v_i^2}$ for sampled v (uniform grid; trapezoidal end weights where integrals are quoted)

2 The residual sees only what the family cannot absorb

The per-run fit holds C_2 at its physical value and minimises over (C_1, C), which is linear least squares on A: the fitted curve is Py and the residual is $\hat{r} = (I - P)y$. The record decomposes as $y = \omega_{\text{nom}} + e_\omega + n$, and $\omega_{\text{nom}} \in V$ exactly (it is $C_1u + (C_1 \cdot 0 + C)\mathbf{1}$ evaluated at the same pinned C_2), so $(I - P)\,\omega_{\text{nom}} = 0$ and

$$\hat{r} \;=\; (I - P)\,(e_\omega + n). \tag{T.1}$$

Hence

$$\text{RMS}(\hat{r}) \;\le\; \text{RMS}\big((I-P)e_\omega\big) + \text{RMS}\big((I-P)n\big), \tag{T.2}$$

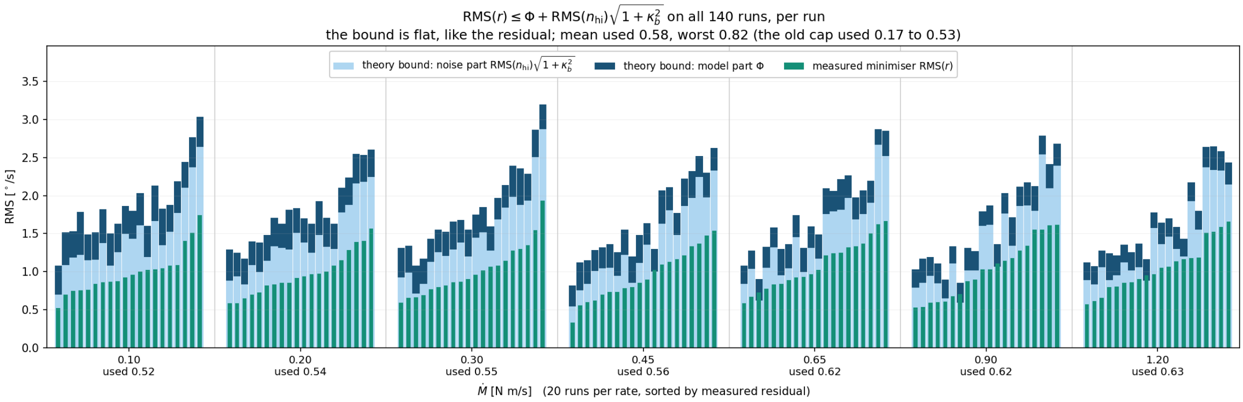

Figure 1: The bound (T.6) against the measured minimiser residual, every run drawn individually: 20 runs per rate, sorted within each rate by the measured residual. Each light bar is the noise part of the bound, $\mathrm{RMS}(n_{\mathrm{hi}})\sqrt{1+\kappa_b^2}$, built from that run's own measured content above 5 Hz; the dark segment on top is the model part Φ; the green bar is the measured residual $\mathrm{RMS}(\hat{r})$. The bound holds on all 140 runs, with a mean used fraction of 0.58 and a worst of 0.82, and it is flat across the rates — like the residual, and unlike the old cap, whose used fraction ran 0.17 to 0.53.

and this is where the old cap gave ground: it bounded the first term by $\mathrm{RMS}(e_\omega) \le \mathrm{RMS}(E)$, the size of the deviation itself. But the minimiser absorbs whatever part of e_ω lies in V, and most of it does. That is not an accident of the data — it is structural. e_ω is built from the kernel columns, and a shifted cosh splits as

$$\cosh\big(C_2(\tau - s)\big) = \cosh(C_2 s)\,\cosh(C_2\tau) - \sinh(C_2 s)\,\sinh(C_2\tau), \tag{T.3}$$

whose first part is $u + \mathbf{1} \in V$ *exactly*. What survives the projection is only the $\sinh(C_2\tau)$ part, weighted by $\sinh(C_2 s)$ — small for impulses early in the window — plus the truncation of the kernel before its impulse time. Norm-weighted over the columns, the family absorbs 68 to 91% of the kernel (79.7% in the median across runs) before any property of ρ is used.

3 The model term Φ

By Duhamel on the deviation dynamics $J_P \ddot{e}_\varphi = W z_{CoM}\, e_\varphi + \rho$ (with $e_\varphi(0) = \dot{e}_\varphi(0) = 0$; $e_\omega = \dot{e}_\varphi$), the sampled deviation is $e_\omega = \sum_j \rho(\tau_j)\, k_j$ to quadrature accuracy. Linearity of $I - P$, the triangle inequality applied pointwise, and $|\rho| \le \bar{\rho}$ give

$$\big|\big[(I-P)\,e_\omega\big]_i\big| \;\le\; \bar{\rho}\sum_j \big|\big[(I-P)\,k_j\big]_i\big| \qquad\Longrightarrow\qquad \mathrm{RMS}\big((I-P)e_\omega\big) \;\le\; \Phi \;=\; \bar{\rho}\,\Big\|\textstyle\sum_j |(I-P)k_j|\Big\|_{\mathrm{RMS}}, \tag{T.4}$$

computable a priori from $(C_2, J_P, \bar{\rho})$ and the sample grid alone. Without the projection the same construction returns exactly the old envelope, $\bar{\rho}\sum_j |k_j(\tau)| = E(\tau)$, so the ratio $\mathrm{RMS}(E)/\Phi$ isolates what absorption is worth: a factor 12 at $\dot{M} = 0.10$ N m/s falling to 4 at 1.20, with $\Phi = 0.40$ down to $0.26°$/s.

Both the representation and the bound were verified on the exact nonlinear tip-over: the kernel sum reconstructs e_ω to 3×10^{-4} °/s, and the realised $\mathrm{RMS}\big((I-P)e_\omega\big)$ is 0.065–0.078°/s across the rates — inside Φ with a factor 4 to 5 to spare, of which about 2 is $\bar{\rho}$ against the realised $|\rho|$ and about 2 is the sign coherence the triangle step discards.

4 The noise term

Projections contract, so $\mathrm{RMS}\big((I-P)n\big) \le \mathrm{RMS}(n)$, and the task is an estimate of $\mathrm{RMS}(n)$ that does not use the residual being bounded. Its high-frequency part needs no estimation: above $f_c = 5$ Hz the family and e_ω are smooth on the window scale and can contribute nothing — and by (T.4) their total contribution anywhere is at most Φ — so n_{hi} is measured directly from the record. The low-frequency part is extended from it by a spectral shape ratio: $\mathrm{RMS}(n) = \mathrm{RMS}(n_{\mathrm{hi}})\sqrt{1+\kappa^2}$ for the in-window κ.

The campaign fixes κ's scale but not its per-run value. On the 113 runs with a quiet stretch of equal window length, κ_q runs 0.15 (p10) to 1.31 (max), median 0.37, and, decisively, κ_q does not predict the in-window ratio run by run (Pearson $r = 0.04$, $p = 0.70$). A per-run noise model is therefore not available, and the bound uses the campaign envelope — measured entirely outside the windows it will be tested on:

$$\kappa_b \;=\; \max_{\text{runs}} \kappa_q \;=\; 1.31, \qquad \sqrt{1+\kappa_b^2} \;=\; 1.65. \tag{T.5}$$

5 The bound, on the campaign

$$\boxed{\mathrm{RMS}(\hat r) \;\le\; \Phi \;+\; \mathrm{RMS}(n_{\mathrm{hi}})\,\sqrt{1+\kappa_b^2}} \tag{T.6}$$

per run, with Φ from (T.4) and the run's own measured n_{hi}. On the 140 runs (means over the 20 at each rate):

$\dot M$ N m/s	Φ °/s	RMS(E) °/s	RMS(E)/Φ	cap (T.6) °/s	old cap °/s	RMS($\hat r$) °/s	used	old used
0.10	0.403	4.876	12	1.87	5.77	0.987	0.53	0.17
0.20	0.364	2.934	8	1.78	3.79	0.964	0.54	0.25
0.30	0.341	2.308	7	1.84	3.22	1.014	0.55	0.32
0.45	0.320	1.892	6	1.67	2.71	0.947	0.57	0.35
0.65	0.295	1.516	5	1.78	2.42	1.084	0.61	0.45
0.90	0.274	1.260	5	1.66	2.10	1.023	0.62	0.49
1.20	0.262	1.120	4	1.68	1.98	1.052	0.63	0.53

It holds on every run — per-run used fraction p10 0.49, median 0.56, p90 0.68, worst 0.82 — it is below the old cap at every rate, and it is *flat*: the cap runs 1.66 to 1.87°/s across a twelvefold change in ramp rate, matching the flat residual, where the old cap fell 5.77 to 1.98.

Little of the remaining width is removable within this form. The binding run fixes the smallest admissible cap factor at 1.28 (against the 1.65 in use), and even at that untenable limit — it is set by the very run it must cover — the mean used fraction only reaches 0.70. The distance from 0.58 to that ceiling is the price of covering the worst spectral shape in the campaign with one constant.

6 What the bound says, and what it cannot

Applying (T.4) to (T.1) two-sidedly gives

$$\big|\,\mathrm{RMS}(\hat r) - \mathrm{RMS}\big((I-P)n\big)\,\big| \;\le\; \Phi, \tag{T.7}$$

so the residual *is* the projected disturbance to within 0.26–0.40°/s. That is the flatness of Fig. 1 read as a theorem: the residual does not depend on $\dot M$ because the disturbance floor does not, and the model error, whose bound does fall with the rate, is capped at a third of the residual's size in the median — 11% of its power. The same statement fixes the limit of this check: a residual test can confirm ρ's contribution is below Φ, but it cannot resolve ρ inside the disturbance floor. The instrument that tests ρ is the forward check of (108), which propagates each perturbation along the measured trajectory and is not noise-limited.

7 Provenance of every constant

C_2, K, J_P are the per-configuration calibration constants, fixed before this analysis. $\bar\rho$ is a priori from the 10° box, as in (93). The kernel, the projector and hence Φ follow from those alone: the model part of (T.6) uses nothing measured in the window. The noise part uses one campaign scalar, $\max \kappa_q = 1.31$, from the quiet stretches only — the bound is built entirely from data outside the windows it is tested on.

One caveat is measured and stated rather than hidden. The in-window spectra are heavier at low frequency than the quiet ones — $s_{\mathrm{med}} = \mathrm{med}(\kappa_{\mathrm{imp}}/\kappa_q) = 1.60$, which is what pivoting on one gear edge adds — so on 5 of the 140 runs the residual exceeds the noise term of (T.6) alone, and it is Φ's margin that covers them. A reader who wants the noise term safe by construction can take $\kappa_b = s_{\mathrm{med}} \times 1.31 = 2.09$: that variant also holds on all 140 runs, at a mean used fraction of 0.44, and prices the in-window enhancement at about 0.6°/s of cap. The tight version is reported as the result and the conservative one as the price of removing its caveat.